
BE 303: Applied Biostatistics

Course Code	BE 303
Course Name	Applied Biostatistics
L-T-P-C	3-0-2-4
Prerequisites	IC252, IC272
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	recommended by the 38th BoA (Jan 2021)
Revised	40th BoA (May 2021) - syllabus revised; 31st Senate (Jun 2021) - renamed from "Biostatistics"

Course Contents

- **Study design, data acquisition, and presentation**

Recognize and give examples of different types of data arising in public health and clinical studies. Types of medical studies and introduction of different study designs (descriptive vs analytical, control groups etc.) and sampling methods (randomization). Risk studies (descriptive, case-control and cohort studies) Measure of important probabilities (incidence, prevalence, sensitivity-specificity-predictivity, morbidity, etc.). Example of study design at the example of a clinical trial for approval process. [10 Lectures]

- **Null hypothesis, Statistical testing**

Why are there statistical tests in medical and biological sciences? Formulation and examples of null hypothesis for medical and bioengineering applications, subsequent decisions, and error types I + II.

Overview of tests: location (e.g., t-test), dispersion (e.g., ANOVA (f-test)), comparison of frequencies (chi-square), variance analysis (e.g., inter- and intra-class variance) and regression analysis (linear and logistic regression (ROC-analysis)). Parametric tests are used only where a normal distribution is assumed. These are the t-test (paired or unpaired), ANOVA (one-way non-repeated, repeated; two-way, three-way), linear regression and Pearson correlation.

Non-parametric tests are used when continuous data are not normally distributed or when dealing with discrete variables. These are chi-squared, Fisher's exact tests, Wilcoxon's matched pairs, Mann-Whitney U-tests, Kruskal-Wallis tests and Spearman rank correlation. Comparison of different tests. Analyze required sample size (calculate the power of a test) and analysis of correct application of a test using specific examples. [15 Lectures]

- **Advanced tools- Introduction to multivariate analysis**

Multivariate analysis is concerned with the interrelationships among several variables. A specific example for the motivation of multivariate analysis will be given. The course includes the following methods: cluster analysis, principal components analysis, factor analysis, discriminant analysis, etc. Designs of Experiments which will include, one-way, two-way ANOVA, MANOVA study design, confounding and standardization. Difference between multiple linear regression and multivariate regression (e.g., using an example of the Framingham heart study) [7 Lectures]

- **Applications of biostatistics, quality, and potential fallacies**

Introduction to survival analysis, estimation of survival curves, and proportional hazards model (e.g., example of life expectancy under different conditions). Analysis of different parametric survival functions.

Quality considerations: Quality of statistical models and quality of data. Dealing with erroneous data, missing values, bias of observer etc.

Demonstration of statistical fallacies due to data (biased sample, inadequate sample size, incomparable objects), erroneous analysis (linear analysis of nonlinearities, biased data selection, misuse of p-values, etc.) and errors or misuse of presentations (misuse of percentages or wrong base for percentages, misuse of means etc. Misuse of graphical representation. [10 Lectures]

Laboratory Modules

Laboratory Modules

- Introduction to R using RStudio
- Analysis of data used in public health with real life examples
- T-test with biological data set
- ANOV A test with biological data set
- Application of Wilcoxon's Signed Rank test
- Application of Mann Whitney U test
- Kruskal-Wallis-test
- Meta-analysis of the disease data
- Equivalence study
- Cross sectional study
- Regression analysis
- Multivariate Methods
- Categorical data analysis

Text Books

1. Abhaya Indrayan and Rajeev Kumar Malhotra, **Medical biostatistics**, Chapman and Hall, 2017.

References

1. Bernard Rosner, **Fundamentals of Biostatistics**, 8th edition, Cengage Learning Inc., 2015.
2. Wassertheil-Smoler, **Biostatistics and Epidemiology: A Primer for Health and Biomedical Professionals**, Springer-Verlag 2014.
3. Shein-Chung Chow, Jen-Pei Liu, **Design and Analysis of Clinical Trials: Concepts and Methodologies**, 3 rd edition, Wiley, 2016.